

Varuna FloodTwin: An Open Urban Flood Digital Twin

Built from Free Satellite Data, and an Honest Measurement of What It Can Predict

Ansh Vivek

Abstract—Cities in India flood every monsoon, and most of them have no calibrated flood model to plan with. We present Varuna FloodTwin, an open pipeline that builds a working flood model of any place on Earth from free satellite data, runs it fast enough to answer questions live in a web page, and serves it for 16 city areas in India. Every stage runs on free computing tiers. Our main claim is not that the model is accurate. It is that we measured what such accuracy numbers are worth, and the answer changes how they should be read. On 150 Sentinel-1 radar scenes across 15 city areas, a baseline with no physics and no rainfall, which simply predicts the cells that were wet in most other storms, scores a Critical Success Index (CSI, the standard flood overlap score) of 0.29 to 0.67. That is higher than any physics-model CSI we can find published for this task, including our own. In all 15 areas it is also higher than the agreement between two radar passes of the same city, which is a ceiling on what any model can be asked to match. Our physics twin scores 0.054, and a learned model reading the same terrain rasters scores 0.288, which ties the no-physics baseline to three decimal places. The learned model’s real value is therefore not accuracy but portability, because it still runs on a city with no radar archive, where the baseline cannot be computed at all. Holding out one entire region at a time it scores 0.147, and a threshold correction that never looks at a test label raises this to 0.175. That correction helps most exactly where the model is weakest, lifting held-out coastal Mumbai from 0.061 to 0.106. Rainfall from free reanalysis contributes nothing at any grouping level, shown four independent ways. Neither model size, from 2.0 to 70.6 million parameters, nor grid resolution, from 60 m to 30 m, moves the result. We then show that planning survives this location uncertainty, because the value of every intervention is measured by re-simulation rather than assumed. A dense storm drain network routed on the real street graph cuts simulated street flooding by 80.9 % in Patna at a lower cost per cubic metre removed than sparse canals, and a metered recharge planner sends 6.3 to 16.5 % of storm runoff into depleted aquifers in six Karnataka towns against 0.7 to 3.3 % on flood plains. An inferred drainage field puts Mumbai’s whole-city drainage ceiling near 5.4 million cubic metres per storm, just above the 4.7 million cubic metres per day the municipal corporation measured pumping in August 2025, which is our first external anchor. Finally, a graph neural network over the street graph amortizes the raster pipeline into a 4.1 ms per-street risk query against 260 ms, transfers to an unseen city at AUC 0.72 to 0.85, and produces evacuation routes recovering 64.6 % of an oracle’s floodwater avoidance at 57 % of its detour.

The author is an independent researcher (e-mail: anshvivek2003@gmail.com).

Code, all per-area data bundles and the live system are public. Repository: <https://github.com/Maverick-Ansh/project-varuna>. Dashboard: <https://project-varuna-iota.vercel.app>. API: <https://anshvivek-varuna-floodtwin.hf.space>.

Index Terms—Urban flooding, digital twin, Sentinel-1, synthetic aperture radar, flood validation, graph neural networks, managed aquifer recharge, open data.

I. INTRODUCTION

PLUVIAL flooding, which means flooding caused by rain falling on the city itself rather than by a river bursting its banks, is one of the most frequent climate impacts in South Asian cities. The planning reality in most Indian municipalities is a drainage map older than the city that grew around it, no calibrated hydraulic model, and no budget to buy one.

A tool that could change decisions in that setting has to meet three conditions. It must be buildable from free and global data for any area, because nobody is going to survey the city for us. It must be cheap enough to run interactively, because planning is a conversation and not a batch job. And it must be honest about what it can and cannot claim, because a confident wrong answer about which street floods is worse than no answer at all.

This paper describes such a system, and it also argues for a particular way of reporting its skill. Fig. 1 is what a planner actually sees. It is a live map of Patna at a chosen rainfall. Blue shading is predicted water depth, arrows point the way water is flowing, orange circles are drain inlets the system placed, purple lines are the storm drain network it designed, and the dark labels name real roads with a depth a person can picture.

Our contributions are the following.

- 1) **A reproducible pipeline from satellite to dashboard on free infrastructure.** A differentiable two-dimensional flood model [1] is built per city from FAB-DEM [2], ESA WorldCover [3] and JRC surface water [4]. A U-Net emulator [5] answers what-if questions in milliseconds. A public dashboard serves rainfall sliders, planners, exposure views and a grounded text brief. It is deployed for 16 city areas.
- 2) **A measurement of what a flood CSI is actually worth,** using trivial baselines that cost nothing to compute and that we think should accompany every CSI reported in this literature.
- 3) **Three corrections against our own earlier published claims,** each forced by an experiment we ran ourselves, and each reported with the number that forced it.

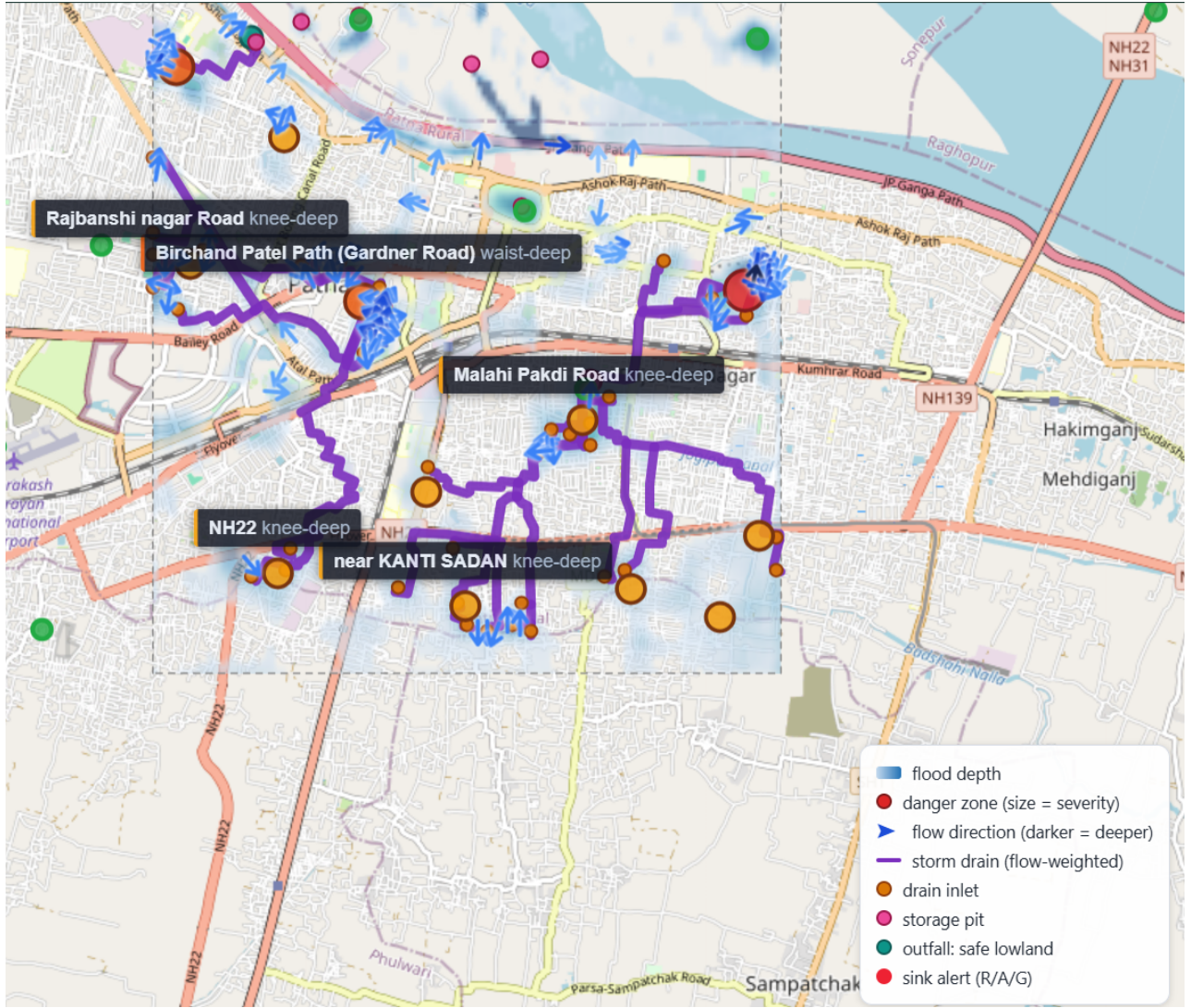


Fig. 1. The live system for Patna at a 100 mm design storm. Blue shading is predicted water depth and the arrows show flow direction, with darker arrows meaning deeper water. Purple lines are the storm drain network the planner designed on the real street graph, and the line gets wider as more water accumulates in that trunk. Orange circles are drain inlets, pink circles are storage pits, and green circles are outfalls placed on safe low ground rather than on the river. Red circles are danger zones sized by severity. The dark labels name real streets and give the depth in words, because “waist-deep” is a decision and “0.94 m” is only a measurement. Everything here is computed from free satellite data and served from a free hosting tier.

- 4) **A transfer result measured the hard way**, holding out an entire region rather than one tile, with a control that rules out the obvious alternative explanation.
- 5) **A coverage result for coastal cities**, which turns a suspected hard limit into a data-collection instruction.
- 6) **A threshold correction that costs nothing** and helps most where the model is worst.
- 7) **Planning surfaces whose value is measured by re-simulation**, covering storm drains, distributed storage and aquifer recharge, with an external anchor against real municipal pumping records.
- 8) **A learned street-level risk model** that amortizes the physics, transfers between cities without retraining, and routes people around water.
- 9) **A reproducibility hazard** in which a fixed random seed made a training failure perfectly repeatable, together with the guard that fixes it.

We have tried to write this paper in plain language. Where a technical term is unavoidable, it is explained the first time it appears.

II. RELATED WORK

Flood physics. Our numerical core is the local inertial shallow water scheme of Bates *et al.* [1], the reduced-complexity formulation that made metre-scale two-dimensional urban flood modelling tractable. Its operational descendant is LISFLOOD-FP [6]. Our contribution is not the scheme. It is a re-implementation in PyTorch so that every simulation is differentiable from end to end, which means we can take the gradient of the flood with respect to the model’s own physical parameters. Recent work moves the same way. Inunda [7] is a near-concurrent GPU-native differentiable solver with essentially the same core, and universal shallow water equations with differentiable programming [8] learn a flow-resistance

closure inside the full equations. Neither ships an emulator, a hosted twin or a planning stack, and we do not claim to predate either.

Flood digital twins and emulators. CLDNet [9] and FlowsDT [10] build AI-driven twins for metropolitan flooding. Google’s Flood Hub work [11] predicts extreme floods in ungauged river basins at global scale, which is the river-flood counterpart of what we do for rain-driven city flooding.

Graph neural networks for flooding. SWE-GNN [12] and its multi-scale successor [13] put hydraulics into message passing on a mesh. FloodGNN-GRU [14] and hydraulics-informed message passing [15] do the same for spatio-temporal prediction, and HydroGAT [16] adds graph attention. Our graph is different in kind. It is the road network itself, so what gets predicted is the flooding of a named street rather than of a mesh cell, and the training labels are generated by our own twin rather than by a commercial solver.

Validation and elevation uncertainty. Wing *et al.* [17] set the standard for validating a continental flood model against observations. Marsh *et al.* [18] measure the vertical accuracy of free global elevation models in flood-prone terrain, which is the error budget driving our own uncertainty analysis in Section X. Assumpção *et al.* [19] and Mazzoleni *et al.* [20] study whether citizen observations can improve flood models, which is the question our participatory layer is built to ask.

Drainage design and recharge. Hesarkazzazi *et al.* [21] generate optimal centralized and decentralized drainage layouts by graph-theoretic optimisation. On the water-supply side, managed aquifer recharge as a joint answer to urban floods and groundwater depletion is studied in [22], and the Underground Taming of Floods for Irrigation concept [23] is its best-known field implementation in the Ganges basin. Our recharge planner differs by metering the water it claims to bank through a re-simulated physical model instead of assuming an infiltration rate.

What is missing from this literature. We could not find a paper in it that reports its CSI beside a trivial baseline computed on the same scenes. Section IV argues that this single omission makes most reported flood CSI values impossible to interpret, including the ones we published ourselves.

III. THE SYSTEM

A. Build layer, run once per city on a free GPU

For any area of interest, Google Earth Engine exports four free global datasets. FABDEM [2] gives ground elevation at 30 m with forests and buildings stripped out, so that the surface is what water actually runs over. ESA WorldCover [3] gives land cover at 10 m, which tells us whether a cell is concrete, soil or vegetation. JRC surface water [4] tells us which cells are permanently wet, so that a lake is not mistaken for a flood. Sentinel-1 radar scenes [24] give observed water on real storm dates, and radar is used because it sees through cloud, which matters when the thing being observed is a monsoon.

The model domain is a square grid at 60 m. Patna-family and Bengaluru bundles use 128^2 cells, which is a window about 7.7 km across, and each Mumbai tile uses 256^2 cells, about 15.4 km. The Mumbai tiles share exact latitude and longitude

edges, so adding them up never double-counts any ground. Those tiles are independent models with closed boundaries, which means water does not flow from one tile into its neighbour and tides are not simulated. The dashboard says so on the page.

The solver is the local inertial shallow water scheme of Bates *et al.* [1], written in PyTorch. Writing it in PyTorch is the point. Every step is differentiable, so we can ask what change in a physical parameter would have changed the flood, and answer it with a gradient rather than by trying values one at a time. Land cover classes map to Manning roughness, which is a number describing how much a surface slows water down, and to infiltration, which is how fast water soaks into the ground. Rainfall forcing comes from Open-Meteo reanalysis and forecast [25].

B. Metered infiltration, and a physics correction that made our floods worse

An earlier version let every pervious cell soak up water at a constant rate for as long as the storm lasted, which is wrong. Real soil fills up. Version 3 meters infiltration against a finite soil storage capacity and caps the rate at the saturated hydraulic conductivity of the soil, which is the fastest water can physically move through it. This change makes our predicted flooding *worse*, by design, and we committed the before and after comparison so the change is auditable. At Patna and 100 mm of rain, water that soaks in falls from 465 449 to 391 525 m³, ponded water rises by 1.4 %, and predicted flooded area goes from 20.63 to 20.78 km². Soil fills to only 2.7 % of its capacity in that storm, which is the useful part of the result. It says the earlier model was not merely mis-parameterised. It was letting a nearly empty sponge absorb water forever.

C. A millisecond emulator, and a reproducibility hazard

A U-Net [5] is trained to imitate the solver so the dashboard can answer what-if questions instantly. Wet cells carry twenty times the weight in the loss, because flooding is rare in space and a plain average error is minimised by predicting no flood anywhere. Validation error across the bundles is 5 to 21 cm and the dose-response is monotone, meaning more rain always produces more water.

Building this exposed a failure worth reporting. Our dataset generator seeds the global random number generator so that storm sampling is reproducible. That also makes the network’s starting weights bit-identical on every rebuild. On the Mumbai Island-City tile, where much of the window is harbour, that particular starting point collapses deterministically. Within one epoch the optimizer pushes the softplus output head so far negative that its gradient vanishes, and the emulator is stuck predicting zero water everywhere. The validation error looks plausible at 16.3 cm, which is exactly the root mean square of the targets, and it stays frozen to the last digit for 40 epochs. A frozen metric is the tell, because it means the gradients are exactly zero.

The important part is that rebuilding cannot fix this. We measured three bit-identical collapses before diagnosing it.

A fixed seed makes failures reproducible too. The shipped trainer now detects a collapsed validation prediction, meaning no cell above the wetness threshold despite wet targets, and retrains from an explicit alternate seed. The first alternate seed turned the worst tile into the best-fitting emulator in the system, at 5.0 cm validation error. We flag the pattern because build pipelines that reseed globally for data reproducibility are common, and a frozen validation metric is an easy and general warning sign.

D. Serve layer, on free CPU hosting

A FastAPI backend serves the committed per-city bundles, including cached OpenStreetMap road graphs [26] and exposure counts, so the deployed service needs no external API access at run time. It re-runs interventions live in seconds. A React and Leaflet front end exposes rainfall what-ifs, ward alerts, drainage, storage and excavation planners, building and road level exposure, validation panels, and a text brief grounded in the bundle’s own numbers. A night lights layer [27] overlays satellite-observed power outages. Everything from build to serve fits inside free tiers, and a new city window takes roughly 15 to 25 minutes to build on a free T4 GPU, including every artifact the server needs.

The system is deployed for 16 areas. These are Patna and two sub-windows, Bengaluru, six Mumbai tiles, and six towns in Karnataka. A seventeenth area, Chennai, exists as a data bundle used for the coastal experiment of Section VII but is not served, which is why the count of deployed areas and the count of areas with radar ground truth are different numbers in this paper.

IV. WHAT A FLOOD CSI IS ACTUALLY WORTH

The most useful thing we can report is not a model score. It is the range a flood score can occupy on this task *before any model exists*.

A. The score, and the three baselines

The Critical Success Index is the standard overlap measure in flood mapping. Take the cells the model says are wet and the cells the radar says are wet. CSI is the number of cells both agree on, divided by the number of cells either one calls wet. It runs from 0 to 1. Three baselines cost nothing and bracket it.

All-wet predicts every cell is flooded. Its CSI is exactly the observed wet fraction, earned with zero knowledge. *Random* scatters the correct number of wet cells uniformly at random. *Climatology* predicts the cells that were wet in most *other* storms of that same area. It uses no rainfall, no physics and no elevation, and it has one threshold which we fix at 0.5 for every area rather than tuning it per area, so it is not fitted to the scenes it is scored on. We also compute a fourth quantity, *SAR versus SAR*, which scores every radar date of an area against every other radar date of the same area. That is not a baseline. It is an empirical ceiling, because it measures how much two radar passes of the same city agree with each other.

B. The result

Table I is the answer. A climatology with one fixed threshold and no other parameter scores 0.29 to 0.67, with a mean of 0.545. That is higher than any CSI we are aware of being reported for a physics-based urban flood model on Sentinel-1, including our own.

It is also higher, *in all fifteen areas*, than the agreement between two Sentinel-1 passes of that same city. Averaging many passes is more self-consistent than any two of them, which is a property of the target and not of a model. This caps what any model can honestly be asked to do here.

The reason the level is so high is that the Sentinel-1 target is dominated by water that has nothing to do with the storm and that the JRC permanent-water mask does not remove. Tidal flat, mangrove, seasonal pond and paddy field are all wet on most dates. CSI on this task has therefore mostly been measuring *can you find standing water*, which our twin never attempted, because the twin predicts only the water the storm adds.

The signature confirms this reading. On Mumbai-Northeast’s one genuine flood, 2026-07-08, with 2937 wet cells against a baseline near 450, the climatology scores its *worst* value of 0.075. It wins the quiet dates and loses the flood.

C. What we recommend

CSI in this literature should always be reported beside all-wet and a leave-one-out climatology computed on the same scenes. Without them the number is uninterpretable. That applies to the numbers in the rest of this paper, and in retrospect it applied to the 0.041 we published earlier.

V. THE TWIN, RE-SCORED, AND TWO CORRECTIONS AGAINST OURSELVES

An earlier version of this work reported that the dynamic twin scores CSI 0.041 against Sentinel-1, that static terrain indices score the same, and concluded that *no* terrain-based router can place ponding in this data regime. Two of those three statements survive re-measurement. The third does not, and we correct it here, because the experiment that refutes it is our own.

A. Protocol

Sentinel-1 water masks for 18 monsoon storms across two areas, Patna with 7 storms from 2023 to 2025 and Mumbai-Northeast with 11 storms from 2025 to 2026, are scored on a common 60 m grid with permanent water masked *on both sides*. Every method, whether physical, index-based or learned, sees the identical grid, mask and target. Every method’s detection threshold is chosen on the *other* storms of the fold and never on the scene being reported. One Patna date, 2024-07-07, has zero observed wet cells and therefore no ground truth, and it contributed a forced 0.0 to every method’s mean in our earlier table. It is dropped. We verified the scoring path by re-deriving the earlier table from it under the earlier

TABLE I

WHAT A CSI IS WORTH ON SENTINEL-1 FLOOD EXTENT, PER AREA, WITH NO MODEL AT ALL. 150 RADAR SCENES ACROSS 15 AREAS, OF WHICH 149 ARE SCORED. CLIMATOLOGY PREDICTS THE CELLS WET IN MOST OTHER STORMS OF THAT AREA, WITH ONE THRESHOLD FIXED AT 0.5 EVERYWHERE. SAR VERSUS SAR SCORES EVERY DATE AGAINST EVERY OTHER DATE OF THE SAME AREA, WHICH IS AN AGREEMENT CEILING BETWEEN TWO RADAR PASSES OF THE SAME CITY. EVERY VALUE IS REGENERATED BY `CSI_NET/BUILD_TRIVIAL.PY`.

area	storms	all-wet	random	climatology	SAR vs SAR
Patna	7	0.054	0.028	0.286	0.214
Bengaluru	10	0.022	0.011	0.577	0.502
Mumbai-South	10	0.017	0.008	0.553	0.464
Mumbai-West	10	0.019	0.010	0.464	0.387
Mumbai-East	10	0.020	0.010	0.598	0.511
Mumbai-North	10	0.024	0.012	0.543	0.459
Mumbai-Northeast	11	0.013	0.007	0.391	0.271
Mumbai-Harbour	11	0.021	0.010	0.572	0.462
Chennai	10	0.021	0.011	0.500	0.383
Doddaballapura	10	0.014	0.007	0.674	0.568
Ramanagara	10	0.009	0.004	0.668	0.586
Chitradurga	10	0.019	0.010	0.671	0.593
Chamarajanagara	10	0.006	0.003	0.522	0.464
Kolar	10	0.030	0.015	0.553	0.458
Chikkaballapur	10	0.045	0.023	0.604	0.523
mean	149	0.022	0.011	0.545	0.456

TABLE II

ALL METHODS ON THE IDENTICAL 18 STORMS, WITH EVERY THRESHOLD CHOSEN OFF THE REPORTED SCENE. THE LEARNED MODEL IS A 7.9 M-PARAMETER U-NET OVER 23 TERRAIN CHANNELS WITH RAINFALL REMOVED, AVERAGED OVER 3 SEEDS. EVERY TERRAIN METHOD IS QUOTED AT ITS MOST FAVOURABLE THRESHOLD RULE. POD IS PROBABILITY OF DETECTION, THE FRACTION OF TRULY WET CELLS FOUND.

method	mean CSI	bias	POD
random	0.014	1.00	n/a
all-wet	0.029	n/a	1.00
static depression depth	0.032	n/a	0.21
TWI (topographic wetness)	0.039	n/a	0.26
HAND-lite (height above drainage)	0.039	n/a	0.90
dynamic twin (antecedent rain)	0.054	n/a	0.29
learned (terrain only)	0.288 ± 0.007	1.41	0.52
climatology (no model, no rain)	0.288	1.29	n/a

protocol, which moved the twin from 0.0412 to 0.0409, TWI from 0.0422 to 0.0429, and HAND-lite from 0.0505 to 0.0504.

We keep this like-for-like table restricted to these two areas deliberately. They are the only areas for which a twin baseline exists to compare against, and mixing in areas where only the learned model can be run would not be a fair head-to-head. The larger 15-area evidence base is used everywhere else in this paper.

B. Correction one: the twin is better than we published

On a fair test set the twin reaches 0.054, not the 0.041 we reported. Dropping the dead scene and adding a second city both help it. It also clears all-wet at 0.029 and every static index, which our Patna-only table did not show. We state this plainly because the correction runs against our own earlier headline in our favour, and a correction that only ever runs one way is not a correction.

C. Correction two: co-location is learnable, so our terrain-fundamental claim was wrong

The same elevation, the same land cover and the same soil rasters, read by a U-Net instead of routed by a solver, reach 0.288 ± 0.007 . That is 5.4 times the twin and 9.9 times all-wet on identical scenes. What fails in this regime is *routing* water over a noisy elevation model, not *predicting* where water goes.

The feature that makes the difference is available to both methods. It is elevation minus its own blurred copy at 2, 8 and 32 cells, which is a measure of whether a spot is locally low. Elevation error is spatially correlated, meaning neighbouring cells are wrong together and by similar amounts, so this relative micro-relief survives a bias that destroys absolute elevation.

D. But the learned model ties the climatology it was meant to beat

The learned model scores 0.2884 and the climatology scores 0.2881. That is the same number to three decimal places, a gap of 0.0003 against a seed standard deviation of 0.0066. On a city that has a Sentinel-1 archive, a neural network over 23 terrain channels is worth nothing over the statement “these cells are usually wet”.

This is the hinge of the paper. The learned model’s contribution is not accuracy. It is portability, which we isolate in Section VI.

E. An independent check with a different instrument

Radar tells us where water was. It does not tell us how deep. To test depth we used crowdsourced reports from the public mumbairflood.in service, run by IIT Bombay’s climate studies group with the municipal corporation. Each report gives a reporter’s height and the fraction of themselves the water reached, so depth is the product of the two.

Of 147 raw reports, 38 carry no depth value, 15 fall outside the region because the app is reachable worldwide and the live feed contains points in Argentina and Australia, and 2 report more than a metre of standing water after under 4 mm of rain, which is not rainfall-driven waterlogging whatever else it is. That leaves 94, of which 85 fall inside a built Mumbai tile and 83 are scored. Personal fields are read and discarded at parse time, and a test pins that they never reach a file. These observations train nothing. They are held for scoring only.

The result is bad for the twin, and it is bad in an informative way. Mean absolute error is 0.207 m and root mean square error is 0.310 m, with a bias of -0.162 m, meaning systematic under-prediction. Against the best trivial baseline, which is simply predicting the observed mean depth everywhere, the skill score is -0.46 . Negative skill means the model is *worse* than predicting one constant number for the whole city. No tile is positive, and the range runs from -0.40 on Mumbai-West to -0.64 on Mumbai-South. Correlation with observed depth is $+0.055$ overall and -0.23 on the genuinely wet sites.

F. Two failure modes, separated

This experiment splits the failure cleanly into a part that is fixable and a part that is not.

Magnitude is a resolution artifact, and it is curable. On wet sites the model under-predicts by a stable factor of 4.45, identical at the 0.15 m and 0.30 m thresholds. That ratio is not mysterious. A person reports the depth where they are standing, and the model reports the average over a 3600 m^2 cell. If the same water sits in a pond of area A inside that cell, then observed over predicted is roughly cell area over A . That implies a ponding footprint of $3600/4.45 \approx 810\text{ m}^2$, a square 28.5 m across. That is a flooded street junction. The model is not wrong about how much water there is. It is spreading it over sixty metres of ground.

Ranking is the real failure, and resolution will not fix it. Correlation with observed depth is zero to negative, and it stays that way across every rainfall window and every neighbourhood size we tried. The model cannot say which street floods worse. This is the same co-location failure the radar work found, now confirmed by an independent data source measuring an independent quantity. Two methods, two datasets, one verdict.

We state the limits of this result rather than leaning on it. There are 83 points from one city in one season, so the negative correlation on wet sites is suggestive and not established. People report water where it is surprising or bad, which is plausibly where the model least expects it, and that reporting bias alone could produce anti-correlation without the model being wrong in the way it appears to be. A point measurement and a cell average are genuinely different quantities, so the 4.45 factor is a consistency argument rather than proof. And a report made hours after the peak is compared against a storm maximum the model is entitled to predict.

VI. THE DEPLOYABLE CLAIM IS TRANSFER, AND IT DEPENDS ON WHAT “UNSEEN” MEANS

If a learned model ties a parameter-free climatology on a city that has a radar archive, then the only reason to prefer the

TABLE III
TRANSFER DEPENDS ON WHAT IS HELD OUT. ALL TERRAIN ONLY, 3 SEEDS, IDENTICAL TRAINING BUDGET. THE REBUILT-STACK CONTROL SEPARATES THE EFFECT OF ADDING AREAS FROM THE EFFECT OF REBUILDING THE FEATURE STACK WITH THE NEW BUILDER’S OWN DEFINITIONS.

held out	areas	mean CSI	$\times$ all-wet
one tile (previously reported)	3	0.094 ± 0.021	3.6
one tile, rebuilt stack control	3	0.096 ± 0.009	3.7
one tile	14	0.270 ± 0.015	12.5
one region	14	0.131 ± 0.005	6.1
one region, with Chennai	15	0.147 ± 0.008	6.9

model is that it runs where the climatology cannot be computed. A city with no Sentinel-1 history has no climatology at all. So the question that matters is how well the model does on ground it has never seen.

We previously reported held-out-city CSI of 0.094 ± 0.021 over three areas and called it a floor rather than an estimate. With eleven further areas, being Bengaluru, the four remaining Mumbai tiles and six Karnataka towns, adding 110 new Sentinel-1 masks, it was indeed a floor. The reason it was so low is that a model trained on two areas has almost nothing to generalise from.

A. The size of the correction depends entirely on how you hold data out

Leaving out one *tile* gives 0.270 ± 0.015 , which is 12.5 times all-wet. That looks like a threefold improvement, and it is misleading. With six Mumbai tiles and seven Karnataka areas in the dataset, holding out Mumbai-West leaves five adjacent Mumbai tiles in training, and holding out Chitradurga leaves six other Karnataka towns that share its climate and its tank morphology. That is generalisation to unseen *ground*, not to an unseen city.

So we group the areas into regions and hold out an entire region at a time. That gives 0.131 ± 0.005 , half the per-tile figure, and it is the number we report as the deployable one. It is still 6.1 times all-wet and 1.4 times our earlier three-area floor, at a fifth of that figure’s seed spread.

B. The control that stops this being overstated

Between the old result and the new one we also rewrote the dataset builder, so the feature stack itself changed. That gives an alternative explanation for the gain which has nothing to do with adding cities. We tested it directly by running the same three original areas on the rebuilt stack. The answer is 0.096 ± 0.009 against the committed 0.094 ± 0.021 . The rebuild moved nothing. The gain is areas.

C. Transfer is not uniform, and that is the operationally important part

Held out entirely, the seven Karnataka areas score 0.213 at 10.3 times all-wet and Patna scores 0.166 at 3.1 times. The six Mumbai tiles score 0.033, which is 1.8 times all-wet and close to predicting nothing at all.

Mumbai’s Sentinel-1 water is tidal flat, creek and mangrove. Neither the Gangetic flood plain nor the Karnataka plateau

TABLE IV

ADDING ONE COASTAL CITY TO THE TRAINING SET. LEAVE-ONE-REGION-OUT, TERRAIN ONLY, 3 SEEDS. EVERY FOLD GAINS THE SAME 10 CHENNAI STORMS, AND ONLY THE COASTAL REGION MOVES.

region held out	14 areas	15 areas	change	× all-wet
Mumbai (coastal, 6 tiles)	0.033	0.061	+84 %	1.8 to 3.3
Chennai (coastal, new)	n/a	0.164	n/a	7.8
Karnataka (inland, 7 areas)	0.213	0.221	+3.6 %	10.6
Patna (inland, 1 area)	0.166	0.154	−7.4 %	2.8
overall	0.131	0.147		6.9

contains a single example of that kind of surface. That admits two readings which fourteen areas cannot separate. Either coastal ground needs coastal training data, or tidal water is not predictable from terrain at all. Section VII separates them.

VII. THE COASTAL LIMIT IS COVERAGE, NOT UNLEARNABILITY

To separate the two readings we built a fifteenth area. Chennai is 1300 km from Mumbai, shares no catchment with it, and is kept as its own region on purpose. That way Chennai’s fold trains on a set containing Mumbai while Mumbai’s fold trains on a set containing Chennai, and one experiment tests both directions at once. Pooling the two into a single “coastal” region would hold both out together and destroy the test.

A. A data-collection bug that nearly hid the answer

Building Chennai exposed a defect worth reporting, because it would have produced a confident wrong conclusion. Our scene selector had the monsoon window hard-coded to 1 June through 15 October. India has two monsoons. Chennai floods in the northeast monsoon, which runs October to December, so the scenes we could even ask for were the dry-season ones.

The first ten masks came from that broken window. After fixing it, the candidate set gained 17 October to December passes and the picture inverted. Northeast monsoon dates average 35.1 mm of three-day antecedent rain against the southwest monsoon’s 14.2 mm, the wettest date goes from 40.3 mm to 180.5 mm, and 8 of the top 10 dates are now October to December. The date 2025-12-01 has a wet fraction of 0.0536, which is five to ten times any other Chennai scene. It is an actual flood, and the first selection could not see it.

The general lesson is that a season window is a scientific parameter and not a configuration detail. Any area whose rain does not arrive between June and October needs its own window, and the code now takes the months through to the satellite query rather than filtering afterwards.

B. Adding one coastal city nearly doubles held-out skill on the other one

Mumbai goes from 0.033 to 0.061. Chennai, scored with Mumbai in training, reaches 7.8 times all-wet, close to inland Karnataka’s 10.6 and nothing like the 1.8 Mumbai managed as the only coast in the dataset. Tidal water is learnable from terrain. The model simply has to have seen a coast.

The experiment carries its own control, which is what makes it worth reporting. Every fold gained the same ten storms, and

TABLE V

THRESHOLD CALIBRATION, TERRAIN ONLY, 3 SEEDS EACH. THE ORACLE-THRESHOLD COLUMN IS THE CSI ACHIEVABLE IF THE BEST SINGLE THRESHOLD WERE CHOSEN WITH KNOWLEDGE OF THE TEST SCENE, SO IT IS AN UPPER BOUND RATHER THAN A METHOD. THE LAST COLUMN IS HOW MUCH OF THAT GAP THE FREE CORRECTION RECOVERS.

held out	global thr.	calibrated	oracle thr.	gap closed
region, 15 areas	0.1474	0.1746	0.1845	73 %
region, 14 areas	0.1305	0.1566	0.1615	84 %
tile, 14 areas	0.2698	0.2542	0.3030	−47 %

only the coastal region moved. The change is +84 % on the coast against +3.6 % and −7.4 % inland. For scale, going from 3 to 14 areas was 4.6 times the storms for +39 % overall. This was +13 % more storms for a doubling of one region.

Two limits we do not overstate. Mumbai at 3.3 times all-wet remains well short of inland performance, so one coastal sibling narrows the gap without closing it. And Chennai’s own figure rests on ten scenes. The direction replicates across three seeds. The magnitude does not yet have tight bounds.

The practitioner’s rule that follows is concrete. Deploy on an unseen inland city freely, and on an unseen coastal city only with at least one coast in training. That is now a measured statement rather than a caution.

Finally, the climatology scores 0.545 on these areas and still wins wherever it can be computed. But it needs the target city’s own Sentinel-1 archive. For a city without one it does not exist, and the learned model remains the only method in Table II that runs at all. That, bounded by the coastal result above, is the claim.

VIII. A FREE THRESHOLD CORRECTION, WORTH MOST WHERE THE MODEL IS WORST

In every experiment in this paper, the model’s ranking quality has run above its achieved CSI. That gap is calibration loss. The model orders cells better than any single global cut-off can exploit. A cell either goes above the threshold or it does not, and if the threshold is in the wrong place then good ordering is wasted.

The fix costs nothing. For each scene, pick the threshold that makes the *predicted* wet fraction match the mean observed wet fraction of the *training* scenes. That prior is a number available without ever looking at the test label, so this is a free correction and not a peek at the answer. Concretely, if the training scenes are 3 % wet on average, take the top 3 % of predicted cells for this scene, whatever the raw probabilities happen to be.

A. The result, and why the two holdout schemes disagree

Table V reports it. On the leave-one-region-out split, which is the deployment question, calibration is worth +0.0272 CSI at 15 areas. That is 3.4 seed standard deviations, and it captures 73 % of the gap between the deployed threshold and an oracle threshold chosen with knowledge of the test scene. It replicates at 14 areas at +0.0261, which is 5.7 seed standard deviations.

On the per-tile split the same correction *hurts*, at −0.0155.

That sign flip is the mechanism, not noise. Hold out one tile and five sibling tiles of the same city stay in training, so

TABLE VI

CALIBRATION BY HELD-OUT REGION, TERRAIN ONLY, 15 AREAS, 3 SEEDS. BIAS IS THE RATIO OF PREDICTED WET CELLS TO OBSERVED WET CELLS AT THE GLOBAL THRESHOLD, SO A VALUE BELOW 1 MEANS THE MODEL IS UNDER-WARNING.

region held out	global	calibrated	change	bias
Mumbai (coastal)	0.0613	0.1061	+73 %	0.78
Chennai (coastal)	0.1642	0.1889	+15 %	0.81
Patna (inland)	0.1537	0.1711	+11 %	0.51
Karnataka (inland)	0.2207	0.2335	+5.8 %	0.71

the training wet-fraction prior already describes the held-out tile well and the swept global threshold is already close to right. Forcing the wet fraction to a prior can then only move it away. Hold out an entire region and the global threshold is genuinely miscalibrated for ground the model has never seen, so matching the prior recovers most of what the ranking was already capable of.

This supersedes an earlier reading of ours. We previously measured +0.0154 from this correction on a three-area per-tile split, called it about one seed standard deviation, and left it as a suggestion. On the split that actually corresponds to deployment, and with five times the data, it is a replicated effect.

B. It helps most exactly where the model is weakest

Table VI breaks it down by region. Every region gains, and the coastal one gains most.

The bias column explains why. At the global threshold the model finds only half to four-fifths of the water that is actually there. Under-warning is the wrong direction for a flood product, because the cost of missing a flooded street is not the same as the cost of falsely flagging a dry one. Ranking degrades far less than CSI does, which is what told us this was calibration rather than a failure of skill.

Put together with the coverage result of Section VII, held-out coastal Mumbai has moved from 0.033, when no coast was in training, to 0.061 once one coast was added, to 0.106 with the free threshold correction applied. That is 3.2 times the starting figure, and it is roughly 5.7 times the all-wet baseline for that region. Neither step required a larger model or a finer grid. One was data coverage and the other was arithmetic.

IX. THREE LEVERS THAT ARE NOT LEVERS

When a model underperforms, the reflex is to make it bigger, give it a finer grid, or feed it more inputs. We tried all three. None of them is the bottleneck here, and we report that because a negative result about the obvious fix saves other people the same weeks.

A. Capacity is not the bottleneck

On held-out floods with terrain only, CSI is 0.221 at 2.0M parameters, 0.232 at 7.9M, 0.221 at 31.4M and 0.219 at 70.6M. That is flat across a 35-fold range and entirely inside seed noise.

B. Resolution is not the bottleneck either

Our standing explanation for poor co-location was that real ponds are around 810m², roughly 28.5m across, and were being scored on 60m cells. Section V is where that 810m² figure comes from, and it is a good explanation for the *magnitude* error. It is not the explanation for the location error. Re-running the whole experiment at 30m quadruples the cell count and changes nothing. The skill multiple over all-wet is 10.4 at 30m against 9.7 at 60m. Raw CSI is not comparable across grids because the base rates differ, which is why the multiple is the column to read.

C. Rainfall is not usable, and it actively hurts

This is the most counter-intuitive result in the paper, so we tested it four ways.

Ablation. Removing the rainfall channels *improves* the model’s ranking quality by 0.124, at about seven seed standard deviations, with the three-seed ranges disjoint.

Permutation control. We shuffled which rainfall vector goes with which storm, preserving every rainfall distribution exactly and destroying only the pairing. The result lands monotonically between real rain and no rain, at 0.184 then 0.272 then 0.313. Real rainfall is strictly the worst of the three inputs. The network fits a rain-to-wetness mapping on the training storms that does not transfer.

Two alternative explanations, tested and refuted. The first was tidal contradiction between the two Mumbai tiles, which predicted that dropping the tidal tile would stop the harm. It did not. The second was failure to extrapolate to larger storms, but rain costs 0.09 CSI within the training distribution too.

The boundary. On an unseen region, rainfall neither helps nor hurts, at 0.1460 with rain against 0.1474 without, replicated at 15 areas. So the claim is not that rainfall is universally harmful. It is that rainfall lets the model take a shortcut on a city it has already seen, and that shortcut does not survive the move to new ground.

D. Why there is nothing in the rainfall to extract

Two properties of the free rainfall archive explain the result. Nine probe points spanning a 15km tile collapse to a single archive cell, so spatial rainfall is not merely unused. It is unavailable.

And the source was not what we previously stated. Open-Meteo’s archive endpoint, queried without an explicit model parameter, returns ECMWF IFS at 9km rather than ERA5. Over the same window and point that is 554.2mm against ERA5’s 1213.4mm, a factor of 2.19 which exceeds any effect our physics calibration ever produced. The twin was never calibrated on ERA5. We now pin the model explicitly, which reproduces every number in this paper exactly. This is the third correction we make against our own earlier statements.

X. WHY THE PHYSICS ROUTER FAILS

Sections V and IX say that terrain contains the information and that the solver cannot use it. This section says why, and

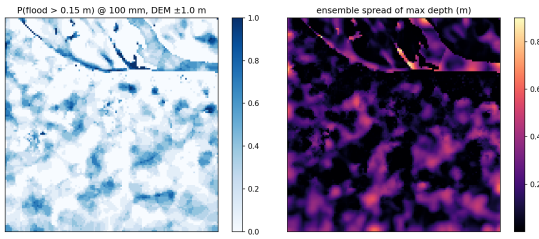


Fig. 2. Elevation ensemble at ± 1 m, showing flood probability over ten members for Patna at 100 mm. The total area is stable but its location moves. Only 5 % of flooded area is common to at least 90 % of members on these flat deltaic plains, against 52 % in hillier Bengaluru.

the answer is a property of the elevation data rather than of the solver.

We re-simulate under a ten-member ensemble of elevation perturbations that are spatially correlated, with a standard deviation of 1 m and a correlation length of five cells, which is FABDEM’s nominal error class [18]. Spatially correlated means neighbouring cells are wrong together rather than independently, which is how real elevation error behaves.

Total flooded area is robust. Patna gives $10.79 \pm 0.35 \text{ km}^2$ at 100 mm. But only 0.58 km^2 , which is 5 %, floods in at least 90 % of the ensemble members. In Bengaluru, with ten times steeper relief, 4.56 of 8.82 km^2 , or 52 %, is robust. Fig. 2 shows it.

Calibrating 16 per-land-class roughness and infiltration multipliers by gradient descent moves held-out CSI only from 0.0483 to 0.0488, with the multipliers staying between 0.94 and 1.14. Roughness and infiltration change *how deep* the water is, not *where* it is. We also ruled out a georeferencing bug, because mirroring the radar mask doubles CSI, which is alarming until you check that the elevation model is provably correctly oriented. The correlation between negative elevation and JRC water is 0.80 unflipped against 0.05 flipped, and the Ganga sits 7.8 m below the domain mean. The flip is a coincidental mirror and is not applied.

The scope of this diagnosis is narrower than we previously claimed. On a flood plain where metre-scale elevation noise exceeds the relief that decides which street ponds, a solver that routes water downhill cannot place it reliably, and its calibration gradient with respect to location is essentially zero. That is why the twin sits at 0.054. It is *not* a proof that the terrain data lacks the information, because a model reading relative micro-relief from the same rasters reaches 0.288. The correct statement is that absolute elevation is too noisy to route over, while relative elevation is not too noisy to learn from.

Two things follow. Reported skill in this regime should always come with both such an ensemble and the trivial baselines of Table I. And planning tools should be judged on decisions that survive this uncertainty, which is what we turn to next.

XI. INFERRING THE DRAINS A CITY WILL NOT GIVE YOU

The largest known physics gap in the twin is that no storm sewer is modelled at all. The model over-floods precisely where working drains exist. Mumbai will not hand out a

drainage map, so the question is whether the drains can be inferred from what a satellite can see.

Two independent validations had already said the twin knows *how much* water and not *where*. Fitting more constants was not going to help, because the 16 per-land-class multipliers of Section X change depth and not location. So instead of more constants we give the model the missing object.

A. Method

After each solver step, water leaves a cell at a fitted rate. This is a sewer and not infiltration, so it never fills the soil column and never counts as recharge. The field is $\text{softplus}(\theta) \times \text{max rate} \times \text{built}$, which forces it positive and restricts it to built land. It is fitted with Adam through the checkpointed simulator on 8 training storms and held out on 3, giving 65 536 parameters per tile instead of 16. There is freedom to move water, priors to keep it physical, and held-out storms to keep it honest.

Three design decisions are worth stating, because each one was a failure first.

Soft-Dice alone cannot train this field. The smooth wet indicator saturates in deeply flooded cells, so the gradient vanishes exactly where the model is most wrong. We added depth hinges whose slope is constant regardless of depth. The first run did 60 iterations and learned nothing.

The loss terms must share units. Expressed as volume fractions of the storm’s rain they are commensurate. With arbitrary weights the fit either drains everything, giving six times more outflow than truth in a synthetic recovery test, or nothing at all.

A minimum-outflow prior charges for every cubic metre drained. This is what makes the volume interpretable, because the fit then picks the *smallest* drainage that explains the observed extent. Every outflow number below is therefore a lower bound and not an estimate.

B. Result one: held-out extent improves by predicting less water

Across four Mumbai tiles the mean held-out CSI goes from 0.0435 to 0.0479, a gain of 10 %, while predicted wet cells fall by 10 %. Per tile the gains are +8 %, +2 %, +3 % and +25 %.

Those are small numbers and we would not report them alone. What makes them worth reporting is the direction of the trade. Section V established that CSI can be bought simply by predicting more water. This is the opposite. Every tile improves while getting drier, which is why the wetness is quoted beside every CSI.

C. Result two: it is the spatial field, not the extra freedom

The obvious objection is that 65 536 parameters will beat 16 parameters at anything. We tested it with a ladder on one tile, holding the loss, the storms and the held-out dates fixed and changing only the parameter count.

A single city-wide drain rate does nothing. The 16-parameter physics calibration makes held-out extent *worse*, because it fits the training storms by turning every multiplier

TABLE VII

THE CONTROLS LADDER ON MUMBAI-EAST. ONLY THE PARAMETER COUNT CHANGES. A SINGLE CITY-WIDE DRAIN RATE DOES NOTHING, THE 16-PARAMETER PHYSICS CALIBRATION MAKES HELD-OUT EXTENT WORSE, AND ONLY THE SPATIAL FIELD IMPROVES IT.

model	params	train	held-out	held-out wet cells
textbook physics	0	0.0428	0.0442	19 691
one uniform drain rate	1	0.0426	0.0439	19 433
per-class Manning/infil.	16	0.0354	0.0402	18 612
per-cell field	65 536	0.0488	0.0478	17 454

up about 1.8 times, which is what “change depth, not location” looks like when you let it try harder. Only the spatial field improves held-out extent. The gain comes from spatial structure and not from parameter count, which is the claim the ladder exists to test.

D. Result three: the decisive test, which it fails

We replayed the 83 quarantined crowdsourced depth reports through both arms with identical machinery, changing only the drain field. Skill against the best baseline moves from -0.918 to -0.921 . Correlation moves from $+0.037$ to $+0.032$. Root mean square error is 0.356 m in both arms. Nothing moves, and the differences are in the fourth decimal, far below meaningful at $n = 83$.

The interpretation is clean. Learning where water *leaves* the surface is not the same as learning where it *goes*. The field is fitted on radar extent, which is a wet or dry mask. That constrains where drainage must exist and says almost nothing about the relative depth of two wet streets. So it improves the quantity it was trained on and leaves the ranking failure untouched. Co-location remains open after three independent attacks, being physics calibration, resolution analysis and now drainage inversion.

E. Result four: what the city pours out, and an external anchor

Integrating the drained volume over each storm answers a question the project could not previously ask. Because the prior selects the smallest drainage consistent with the radar, these are lower bounds.

On the complete six-tile city with design storms, inferred surface outflow is at least 3.48 million m^3 at 25 mm, 4.63 at 50 mm, 5.21 at 100 mm and 5.41 at 200 mm. Flooded land over the same range goes from 70.7 to 241.5 km^2 .

The outflow saturates. Eight times the rain buys 56% more drained water, and the last doubling from 100 to 200 mm buys 4% . The field is rate limited. Once the streets are wet for the whole storm, extra rainfall cannot leave any faster, so it ponds. Note how differently the two columns behave. Drained volume flattens while flooded area keeps climbing. That is the mechanism. Beyond the drainage ceiling, additional rain has nowhere to go but outward across the ground. This is a concrete and falsifiable statement about Mumbai’s drainage limit, at roughly 5.4 million cubic metres per storm however hard it rains.

We can now check it against a real municipal measurement. The six municipal stormwater pumping stations are

rated at 258 m^3/s combined, a theoretical ceiling of 22.3 million m^3 per day with every pump at full rate. More useful than the rating is a measurement. The Brihanmumbai Municipal Corporation reported pumping $16\,451.55$ million litres, which is 16.45 million m^3 , between 16 and 19 August 2025 through those six stations. That is about 4.7 million m^3 per day.

Our inferred ceiling of at least 5.41 million m^3 per storm sits just above that daily pumped volume, which is the right side to sit on. Our number is total surface outflow, and Mumbai has 186 outfalls, most of them gravity fed to the sea at low tide. The municipal figure counts only what six pump houses lifted. A total drainage estimate should exceed pump-only throughput, and it does, by a modest factor rather than an implausible one.

This is a consistency check and not yet a validation, for two reasons we state plainly. Our number is a lower bound by construction, so agreement in magnitude cannot be pushed further than “not contradicted”. And the time bases differ, because ours is per storm and theirs is per day over a three and a half day wet spell. Closing it properly needs the per-event pumped volume for a specific storm that we also model, and two dates in our mask set qualify.

Design capacity is the wrong quantity to compare against, incidentally. The city’s drainage master plan is specified as a rate over an area, at 25 mm/h existing and 50 mm/h designed, which converts to roughly 10.9 million m^3 per hour over the 437 km^2 of the municipal area and so exceeds our per-storm figure by two orders of magnitude. That comparison says nothing, because only 28 of 58 project phases were complete as of 2019 and the rate is nominal. The pumped volume is the number with teeth. Inferred capacities themselves are physically plausible for urban drainage, averaging 0.3 to 3.0 mm/h over built land and peaking at 4 to 50 mm/h.

XII. PLANNING THAT SURVIVES LOCATION UNCERTAINTY

Section X says we cannot reliably name the street that ponds first. Planning is still possible, because the quantities that drive a plan are not the quantity that fails. Every intervention below is evaluated the same way. Modify the terrain, re-simulate the design storm at 100 mm over 24 h, and measure flooding on built-up cells, *excluding* the intervention’s own cells from both the before and the after mask. Moving water onto a drain does not count as removing it. The excluded fraction is reported and stays below a few percent of built cells.

A. A storm drain spiderweb routed on the real street graph

Sparse worst-basin-to-nearest-outfall canals are how our earlier planner worked, and anecdotally how much real retrofitting works. Two to five least-cost corridors over the elevation grid cut 20.8% of street flooding in Patna and 25.7% in Bengaluru. Preferring road cells and avoiding buildings *improved* the Bengaluru cut threefold, which is a useful fact on its own. Streets trace drainage valleys in dense terrain, so buildability and hydraulics point the same way.

Scaling that idea up, we route a dense network on the actual street graph. The area’s OpenStreetMap roads [26] form a graph whose nodes carry elevations, with 42.8 k nodes and

TABLE VIII

MEASURED INTERVENTION LADDER FOR PATNA AT 100 MM, ALL RE-SIMULATED. COSTS USE INDICATIVE INDIAN CIVIL-WORKS RATES OF INR 9 000 PER METRE OF DRAIN, INR 300 PER m^3 OF EXCAVATION AND INR 6 000 PER m^3 OF CONCRETE STORAGE. THE DENSE STREET-ROUTED NETWORK BEATS SPARSE CANALS ON BOTH AXES.

intervention	cut	flood removed	cost per m^3
excavation, 8 sites	4.1 %	0.06 Mm^3	INR 758
detention pits only, 8 sites	7.8 %	0.11 Mm^3	n/a
sparse canals and pits, 3.2 km	20.2 to 20.8 %	0.28 Mm^3	INR 1 309
spiderweb, 40 inlets, 44.5 km	80.9 %	0.87 Mm^3	INR 850
distributed storage, 727 basins	30 %	0.43 Mm^3	INR 9 200

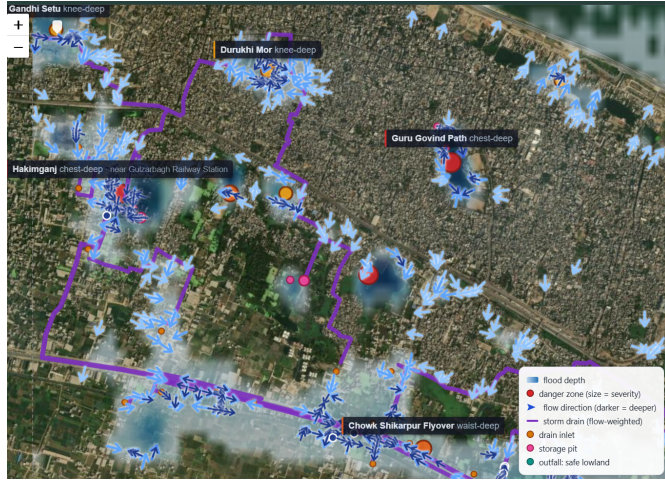


Fig. 3. The designed drain network over satellite imagery for eastern Patna. Purple lines are the planned trunks, orange circles are inlets, and the arrows are the simulated flow field. The labels are real road names carrying a predicted depth in words. The network is not drawn by hand. It is the union of gravity-cheapest exit paths from 40 basin inlets to safe outfalls.

6.8 k ways for Patna. Edge cost is street length plus an uphill penalty of 2 400 m per metre of climb. One reverse multi-source Dijkstra search from all outfalls labels every street with its gravity-cheapest exit. About 40 inlets are placed at the deepest cell of each flood basin, and their exit chains trace out a forest with shared trunks that accumulate flow.

The outfalls encode a safety principle. *Never the river*, because during a flood the river runs high and a drain into it runs backwards. Instead we use the lowest *safe* ground, meaning non-built land buffered away from buildings and from the river corridor, plus downhill exits from the domain and detention pits. Pits are handicapped by 2 km of equivalent cost so that carrying water away wins wherever gravity allows it. The chosen streets are then carved into the elevation as 2 m channels with strictly descending beds, shared trunks take the deepest bed, and the storm is re-simulated.

Results. Across the eight built areas the spiderweb cuts street flooding by 80.9 % in Patna, 89.4 % in Patna-East, 80.3 % in Patna-West, 72.6 % in Bengaluru, and 63.0, 50.2, 39.2 and 32.5 % on the Mumbai South, East, West and North tiles, against 21 to 26 % for the sparse plans. Dose-response over 25 to 200 mm is monotone.

Density improves the *economics* as well as the effectiveness. Patna costs INR 850 per cubic metre removed against

INR 1 309 for the sparse plan, and Bengaluru costs INR 1 121, because shared trunks spread their length across many basins. The lower Mumbai cuts are informative rather than disappointing. Those tiles are four times larger, far denser and partly harbour, so a fixed budget of about 40 inlets covers a smaller share of the flooding.

B. Buildable and phased distributed storage

The storage planner answers a different question. How many detention tanks, and where, to cut the flood by a target percentage? Candidate cells are the dynamic local minima of the simulated flood, ranked by pooled depth. Each site is sized to the water it actually holds, and the cut is re-measured by simulation rather than assumed.

The important word is buildable. Building footprints and permanent water are excluded from candidacy, because you cannot dig a tank under an occupied house or in the harbour. Under-street detention stays allowed, which mirrors how Indian cities actually retrofit. Buildability is not free on flats. Excluding building-footprint minima removes some of the deepest candidates and the reachable ladder drops accordingly. We report that rather than quietly siting tanks inside houses.

Each area ships an explicit site list of up to 1 500 sites with coordinates, per-site volume and an under-street flag, where 63 to 87 % of sites land under streets, together with a phase ladder costed at indicative rates. Patna reaches a measured 10 % cut with 193 sites at about INR 165 crore. Mumbai-West needs 493 sites and about INR 825 crore for the same 10 %. Dense tiles reach 10 % and no further, which is an honest ladder rather than an aspirational one.

The planner has never seen a flood report or a news archive. Its top-ranked sites nonetheless land on each city's chronically flooded places, including Koramangala in Bengaluru and the Mithi at Kalina in Mumbai-West. Fig. 4 shows the Mumbai island city network concentrating on the Parel, Dadar and Byculla belt, which is the Hindmata neighbourhood where the municipal corporation actually built its underground detention tanks. That coincidence is a qualitative check. Deep simulated minima on buildable ground fall where flooding actually hurts.

C. Why we trust these decisions given Section X

The quantities that drive the plans are the ones the elevation ensemble shows to be robust. Those are total flooded volume, the existence of a basin, the approximate ordering of depths, and the downhill topology. They are not the exact street that ponds first. The planner's outputs are also portfolios, made of many inlets and many small basins, whose value degrades gracefully if any single pond turns out to be somewhere else. We therefore frame the outputs as relative planning guidance, meaning which strategy, at what density, and where to start, rather than as certified absolute depths.

XIII. FROM FLOOD TO AQUIFER: METERED RECHARGE

The water that floods an Indian city in July is water the same region is short of in April. Managed aquifer recharge is the idea of deliberately putting storm water underground instead

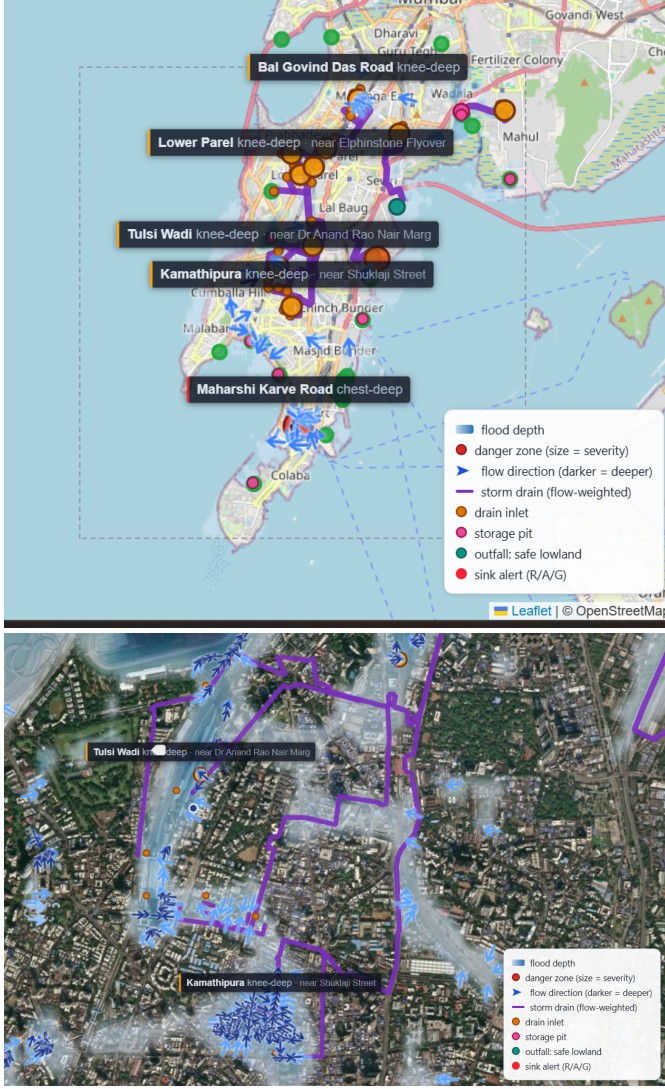


Fig. 4. Mumbai island city. **Top**: the planned network for the whole tile, with labelled streets at knee and chest depth through Lower Parel, Kamathipura and Tulsi Wadi. Green circles are safe-lowland outfalls and pink circles are storage pits. **Bottom**: the same plan over satellite imagery. The trunks follow the street grid because that is where gravity and buildability agree.

of letting it run off, and it is well studied [22], [23]. What is usually missing is a measurement. Most recharge plans assume an infiltration rate and multiply.

We can do better, because we have a physical model that already meters infiltration against a finite soil capacity, as described in Section III. So we can build recharge structures into the model, re-simulate the storm, and read off how much water actually went underground rather than assuming it.

A. Tier A, a state-wide screen

Choosing where to even build a twin is itself a decision, and doing it by intuition would undo the point. We screen an entire state first. For each of Karnataka's 27 district units we combine four free datasets. Central Ground Water Board stage-of-extraction gives the annual ratio of water taken out to water going in, so above 100 % means the aquifer is being mined. A GRACE-assimilated storage trend gives the direction

TABLE IX

METERED RECHARGE AT A 100 MM STORM. RECHARGE IS WATER THAT ENTERED THE SOIL COLUMN IN THE RE-SIMULATED MODEL, NOT AN ASSUMED RATE. THE FLOOD CUT COLUMN IS THE MEASURED REDUCTION IN STREET FLOODING FROM THE SAME STRUCTURES. THE PLATEAU TOWNS BANK AN ORDER OF MAGNITUDE MORE OF THEIR RUNOFF THAN THE FLOOD PLAIN AND THE COAST DO.

area	sites	recharge (m ³)	of runoff	flood cut
<i>Karnataka plateau</i>				
Chikkaballapur	4 323	607 172	16.5 %	62.7 %
Kolar	4 875	667 548	14.1 %	46.9 %
Dodballapur	4 122	503 986	10.0 %	85.4 %
Chamarajanagara	5 215	418 956	8.1 %	88.2 %
Chitradurga	3 850	320 138	8.0 %	49.2 %
Ramanagara	3 119	322 204	6.3 %	76.8 %
<i>Flood plain and coast</i>				
Patna-East	3 159	102 013	3.3 %	38.4 %
Patna-West	3 277	139 793	2.6 %	22.4 %
Bengaluru	986	74 196	1.6 %	4.3 %
Mumbai-North	3 654	114 711	1.1 %	6.2 %
Patna	1 188	32 714	0.7 %	9.1 %
Mumbai-East	2 693	21 171	0.2 %	14.4 %
Mumbai-West	1 573	16 117	0.2 %	7.6 %
Mumbai-South	507	-534	0.0 %	3.7 %

of change. HydroSHEDS flow accumulation says where runoff concentrates, because recharge needs water to arrive. SoilGrids conductivity and WorldCover pervious fraction say whether the ground can take it. The four terms are weighted 0.45 for need, 0.25 for water, 0.15 for soil and 0.15 for land.

The screen ranks Bangalore Rural first at 111 % extraction, then Chitradurga at 144 %, Chamarajanagara at 96 %, Kolar at 180 % and Bangalore Urban at 187 %. All are officially over-exploited or critical.

Two honesty notes travel with this screen. The storage trend input turns out to be effectively constant across the state, so it contributes nothing to the ranking, and we say so in the artifact rather than stretching a flat input into apparent signal. And a screen is not a siting. It can say that a district is depleted *and* has concentrated runoff *and* has pervious ground. It cannot say build here. Only a re-simulated twin earns a claim in cubic metres.

B. Tier B, twins for six towns, and the measurement

We then built full 60m twins for six Karnataka towns and ran the recharge planner in each, along with the flood-plain and coastal areas already built. The planner places recharge structures on pervious ground where water collects, sizes each one, re-simulates, and reports the water that crossed into the soil column.

Table IX is the result, and the split is the point. The six Karnataka towns send 6.3 to 16.5 % of a 100 mm storm's runoff into the aquifer. The flood plain and the coast manage 0.7 to 3.3 %, and Mumbai-South is indistinguishable from zero. That gap is exactly what the screen predicted and it has a physical reason. On a flood plain the water table is already close to the surface and the soil column is nearly full, so there is nowhere to put the water. On the Karnataka plateau the aquifer is mined, hard rock sits under thin soil, and the storage the water is being invited into is real and empty.

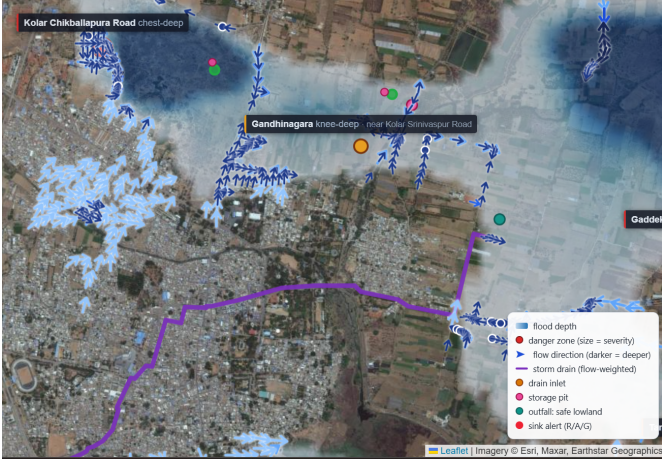


Fig. 5. Kolar, one of the six Karnataka towns, at a design storm. Kolar district runs at 180% groundwater extraction, meaning it takes out nearly twice what goes in each year. Blue is predicted water, arrows are flow, the purple line is the planned trunk to a safe outfall in green, and the pink circles are storage. Water that currently floods Kolar Chikballapura Road is water the aquifer under it needs.

The same structures also cut flooding, by 46.9 to 88.2% in those towns, so this is not a trade between flood control and water supply. In the places where recharge works, it is the same intervention.

Fig. 5 shows Kolar, the second-ranked town, with its flood field and the planned network. Phasing it is costed the same way as storage. Kolar reaches a 10% flood cut with 3296 structures holding 6.94 million m^3 , at an indicative INR 2 500 per m^3 and about INR 1 735 crore.

C. What this result does not license

Two caveats sit on top of these numbers and both are load-bearing.

The recharge *volumes* come from the twin’s metered infiltration and depend only on the physics and the terrain. The site *rankings* additionally use local groundwater level data, and the well files currently shipped are sample placeholders rather than real measurements, with the nearest genuine station far outside the area. So the volumes stand and the within-town ordering of sites should be treated as provisional until Central Ground Water Board station records are attached. We ship that distinction in the artifact instead of hiding it.

Recharge also has a water quality dimension that a flood model cannot see. Putting urban runoff underground puts whatever is in that runoff underground too, including sewage and heavy metals. Parts of Bihar’s shallow alluvium carry arsenic, and Karnataka’s hard-rock aquifers carry fluoride. Any real structure needs geotechnical and water-quality screening before construction. This planner prioritises. It does not certify.

XIV. LEARNED STREET-LEVEL RISK, AND ROUTING PEOPLE AROUND WATER

Both planning surfaces above route on graphs with hand-set costs. We now replace the hand-set cost with a learned one,

TABLE X

FLOODGNN PER AREA. EDGE AUC IS MEASURED ON HELD-OUT STORM SIZES. AUC IS THE PROBABILITY THAT A RANDOMLY CHOSEN FLOODED STREET IS RANKED ABOVE A RANDOMLY CHOSEN DRY ONE, SO 0.5 IS CHANCE AND 1.0 IS PERFECT. THE LAST COLUMNS COMPARE A FULL-GRAPH RISK QUERY AGAINST THE RASTER PIPELINE IT REPLACES, ON THE SAME HARDWARE.

area	nodes	edge AUC	GNN	emulator	speed-up
Patna	37 017	0.906	4.1 ms	260.2 ms	63×
Patna-East	13 590	0.942	3.9 ms	110.9 ms	28×
Patna-West	31 961	0.890	4.1 ms	216.0 ms	53×
Bengaluru	31 675	0.916	5.0 ms	238.4 ms	48×

and in doing so answer a different question. A planner asks where to build. A person standing in the rain asks which way to walk.

A. FloodGNN

FloodGNN is an edge-conditioned message-passing network [28] over the OpenStreetMap street graph, with four rounds of message passing, 96-dimensional states and about 439k parameters. Message passing means each street repeatedly exchanges information with the streets it connects to, so a street can learn that it is at the bottom of a basin even though nothing in its own features says so. The network is conditioned on rainfall by FiLM [29], which lets one model cover a continuous range of storm sizes instead of one model per storm.

Node features are deliberately local and standardised per graph. They are elevation anomaly, pit-ness, grade, flow accumulation, permanent water, radar waterlogging frequency, and building and road density. Nothing in them identifies the city, which is the property that makes transfer possible at all.

Supervision is generated by the twin itself. The emulator’s depth fields over a 20 to 240 mm sweep are sampled at street nodes and edges, and the drain planner’s conveyed volume is sampled at three configurations. That is about nine seconds of label generation per city on a CPU. Wet targets carry the same twenty-fold loss weight as the emulator, validation holds out *storm sizes* rather than nodes, and the shipped checkpoint is the best-validation epoch.

B. Risk, and what the message passing buys

Table X gives per-street flood risk at any rainfall in a single forward pass, at 0.89 to 0.94 AUC on held-out storm sizes, in about 4 ms against 260 ms for the raster pipeline.

An ablation says the graph structure is earning its place rather than decorating it. Replacing message passing with a per-node multilayer perceptron that sees the identical features drops Patna from 0.906 to 0.866, Patna-East from 0.942 to 0.908, Patna-West from 0.890 to 0.879 and Bengaluru from 0.916 to 0.888. The gain is monotone in the number of rounds, at 0.866 with none, 0.890 with two and 0.906 with four on Patna. Neighbourhood structure carries signal beyond what purely local topography explains, which is exactly what a basin is.

We also report what does not work. The flow head recovers the drain planner’s trunk ranking only weakly, with Spearman correlation of 0.14 to 0.24. That is usable for suggesting

TABLE XI

FLOOD-AWARE ROUTING, 150 ROUTES PER AREA, ALL JUDGED ON ORACLE DEPTHS. WET METRES IS THE FLOODED STREET DISTANCE AN AVERAGE ROUTE CROSSES. DETOUR IS THE EXTRA LENGTH AGAINST THE SHORTEST PATH. AVOIDANCE RECOVERED IS HOW MUCH OF THE ORACLE’S IMPROVEMENT THE GNN CAPTURES.

area	shortest	GNN	oracle	recovered
Patna	2 297 m	1 570 m	1 172 m	64.6 %
Patna-East	585 m	406 m	369 m	82.9 %
Patna-West	2 024 m	1 727 m	1 349 m	44.0 %
Bengaluru	1 888 m	1 172 m	1 333 m	129 %

corridors in the dashboard, but trunk placement remains the planner’s job. The network predicts *wetness* well and *flow direction* poorly, and we report the number rather than the claim.

C. Zero-shot transfer between cities

Trained with Bengaluru held out entirely, which makes the training set all flat, FloodGNN ranks flooded Bengaluru streets at 0.721 AUC without ever seeing the city, against 0.916 when Bengaluru is in training. The reverse direction is stronger. Held-out Patna scores 0.847 against 0.906, because a mixed flat and hilly training set already covers Patna’s regime.

This is the learned counterpart of Section VI. Street-level flood knowledge is substantially city-transferable even where cell-level location is not, and it is hardest going into a terrain type absent from training, which is the same lesson the coastal experiment taught at the raster level.

D. Routing

Given per-street risk, routing is a shortest-path problem with an inflated cost, so that crossing deep water is expensive but not forbidden. We evaluate on 150 random origin and destination pairs per area at 140 mm, judging every route on the true depths so that no method is scored on its own beliefs. Three routers compete. The flood-ignorant shortest path, FloodGNN, and an oracle that reads the emulator directly.

On Patna the learned route crosses 1 570 m of flooded street against 2 297 m for the flood-ignorant path and 1 172 m for the oracle. That recovers 64.6 % of the oracle’s improvement while taking a 21.2 % detour against the oracle’s 37.1 %. In other words it buys roughly two-thirds of the benefit for a little over half the walking.

On Bengaluru the learned router beats the oracle. It crosses 1 172 m against the oracle’s 1 333 m. This is not a mistake and it is worth explaining, because it looks like one. The oracle is greedy on true depth, minimising a risk-inflated cost step by step, and that is not the same objective as minimising wet metres. The GNN’s risk field is smoother, having been produced by message passing that averages over neighbourhoods, and a smoother field finds a better global path. An oracle is only optimal for the objective it actually optimises.



Fig. 6. Flood-aware routing in the deployed system. The purple line is the recommended route, orange circles are street nodes the router refuses to cross, the arrows are the predicted flow field, and the label names the specific road that is impassable. The full-graph risk query behind this view takes about 4 ms, which is why the route updates as the rainfall slider moves.

E. Honesty and deployment

FloodGNN distills the twin, so it inherits the twin’s biases. Its claims are amortization, routing quality *under* the twin’s flood fields, and transfer. They are not co-location skill, which Section V shows the twin itself largely lacks and which a model learned directly from terrain does have. Supervising directly on radar-observed street water is the natural next step and we have not done it.

The 1.8 MB checkpoint ships inside the dashboard image. The routing endpoint answers from the GNN where a checkpoint exists and from the raster emulator otherwise, and it reports which backend answered.

XV. THE DEPLOYED SYSTEM AND ITS LEARNING GATE

Everything above is computed from satellites and simulation. The people being flooded know things neither can see. The deployed system closes that loop under strict honesty constraints, and we report the loop’s status accurately, which is that its machinery works and its evidence does not yet exist.

A. Citizen reports as data

Dashboard users drop a pin with a water level they can actually judge, being ankle, knee, waist or chest, mapped to depth bands with explicit tolerances of ± 0.15 to 0.35 m, plus an optional note. This is why the labels in Fig. 1 are words rather than numbers. A person can confirm or contradict “knee-deep”. Nobody can confirm 0.47 m.

Reports are validated against the area’s model window and rate-limited by salted address hash, with five per hour, a per-cell cool-down, a global daily cap and a honeypot field for bots. They are held in memory for the live map and mirrored by a background scheduler to a public dataset repository, which doubles as durable storage across free-tier restarts. No personal data is served, and the hashes never leave the backend.

B. From reports to supervision

A report is only a label together with the storm that caused it. The nightly job groups reports by calendar day in local time, tags each day with the rainfall that actually fell, and drops dry days below 5 mm. A knee-deep report without rain is far more likely to be mischief or a burst pipe than a storm signal. Same-cell same-day reports collapse to their median, so one street corner cannot dominate.

C. The reward gate

Citizen reports carry no dry labels, so a model that predicts deep water everywhere would score perfectly on wet points. That is the same failure mode as the emulator’s sparse targets, now in the field, and it is why the loop needs a gate rather than a learning rate.

A candidate emulator is fine-tuned with a band-tolerant loss on report points, meaning zero gradient inside the reported band, anchored by a simulation replay buffer in every batch. It ships only if all four of the following hold. There are at least 15 held-out points from at least two distinct storm days, held out *by day* because points within a day are correlated. Held-out point error improves by at least 5%. Replay error degrades by at most 10%, which is the all-wet guard. And it wins at least 70% of bootstrap resamples of the held-out errors.

Every decision, accepted or held, with all four numbers, is appended to a public learning log surfaced on the dashboard, so the loop is auditable by the people whose reports feed it.

Status, stated plainly. The loop is deployed and the gate is tested offline. The refresh half, which pulls weather and recomputes outlooks, runs end to end. The learning half has never fired, because the reports dataset is empty. Running it nightly today would produce a log of nights with zero model updates, which is the correct behaviour and not a result. We therefore claim a mechanism and an audit trail here, and no learning outcome. The first accepted or correctly refused field update remains the most interesting experiment we have not been able to run.

D. Live context layers

The same discipline applies to the ambient layers. Live forecast alerts recompute the outlook read-only and never mutate committed artifacts. Advisories are generated in English, Hindi and Marathi from only the system’s own numbers, with a deterministic template fallback so the panel works with no language model at all. A Black Marble night lights overlay [27] marks built-up cells whose latest good-quality night radiance dropped more than 50% below their 90-day median, which is a proxy for a power outage.

One trap there is worth recording for anyone building the same layer. During monsoon cloud cover a fully quality-masked composite downloads as *zero radiance*, which reads naively as a city-wide blackout. Missing data must be treated as unknown and not as dark.

XVI. LIMITATIONS

Resolution. The hydraulics run at 60 m, which is coarser than the streets they describe. Carved drains are full 60 m cells, so absolute excavation volumes are resolution artifacts even though the per-metre drain costing is realistic. Section V measures the consequence directly. Depth is diluted by a factor of 4.45, implying a real ponding footprint near 28.5 m across. A finer grid is the single clearest next step, and it has a quantitative target.

Depths are uncalibrated, and Section X shows why per-class physics calibration cannot fix location inside the solver, even though a learned model on the same rasters can.

The evidence base is uneven. The like-for-like comparison of Table II rests on 18 storms across two areas, because those are the only areas with a twin baseline to compare against. Everything after it rests on 15 areas and 150 scenes. We consider re-measuring the like-for-like table across all 15 areas the most valuable single experiment we have not run.

Mumbai has regime-specific caveats. The tiles are independent closed-boundary models, so flow across the seams is not simulated. Tidal and storm-surge coupling, often the binding constraint for Mumbai outfalls, is absent, and rain falling on harbour cells ponds harmlessly inside the model instead of exchanging with the sea. Mumbai-Harbour is excluded from headline validation because its observed water is almost entirely persistent, with a storm increment of exactly zero on 8 of its 11 dates, so it measures a different hazard than urban rain flooding. A calendar-derived tide phase does not explain it, with a permutation test at $p = 0.47$, and a real tide gauge remains untested.

The coastal result rests on two cities. One coastal sibling raises held-out Mumbai by 84% without closing the gap to inland performance, and Chennai’s own figure rests on ten scenes. Whether coastal skill keeps climbing with a third coast or plateaus is unresolved, and it is a cheap experiment.

Recharge site rankings use placeholder well data, as stated in Section XIII, while the recharge volumes come from the twin’s metered infiltration and do not depend on it.

Costs are indicative. Phase costs use a single civil-works rate. Land, utilities and operations are out of scope, so the ladders rank strategies rather than price projects.

The learning loop is unexercised. Its gate is deployed and tested offline, and no field report has ever reached it.

Routing is evaluated on synthetic trips. The 150 origin and destination pairs per area are random, not observed journeys, and the oracle they are measured against is greedy rather than optimal.

What we would do next. A finer national elevation model, assimilating mapped drain networks where cities have them, sub-grid street conveyance, supervising FloodGNN directly on radar-observed street water to remove the “distills the twin” caveat, tidal boundaries for coastal tiles, a third coastal city, and validating intervention *deltas* rather than state against before and after radar where a city has actually built drainage.

XVII. CONCLUSION

We built a working urban flood twin for 16 Indian city areas out of free satellite data, on free computing, and put it on the

public internet. That part is engineering. The part we think matters more is what happened when we measured it properly.

A flood overlap score on this task is uninterpretable on its own. A baseline with no physics, no rainfall and one fixed parameter scores 0.29 to 0.67 across 15 areas, beating every physics-model score we can find published, and beating the agreement between two radar passes of the same city in all fifteen. Any CSI in this literature, ours included, should be read next to those numbers or not read at all.

Once that scale is fixed, the rest of the results change shape. Our physics twin scores 0.054 and a learned model on the identical rasters scores 0.288, which refutes our own earlier claim that terrain fundamentally lacks the information. But that same learned model ties the parameter-free baseline exactly, so its value is not accuracy. Its value is that it runs on cities with no radar history at all, and held out one whole region at a time that is worth 0.147, rising to 0.175 with a threshold correction that costs nothing and never looks at a test label. The one place it nearly fails is a coast it has not seen, and we showed that is a coverage problem rather than a hard limit, because adding a single coastal city lifted held-out coastal skill by 84 %.

Three things that ought to have helped did not. Model size across a 35-fold range did nothing. Grid resolution from 60 to 30 m did nothing. Rainfall from free reanalysis did nothing at best and actively hurt at worst, shown four independent ways with two alternative explanations refuted.

Meanwhile planning works despite the location uncertainty, because we measure every intervention by re-simulation instead of assuming it. A dense street-routed drain network cuts simulated street flooding by 80.9 % in Patna and costs less per cubic metre than the sparse alternative. A metered recharge planner banks 6.3 to 16.5 % of a storm’s runoff in six depleted Karnataka towns against 0.7 to 3.3 % on the flood plain, and cuts flooding at the same time. An inferred drainage field puts a falsifiable ceiling on Mumbai’s whole-city drainage near 5.4 million cubic metres per storm, which sits on the correct side of the 4.7 million cubic metres per day the city measured itself pumping. And a graph network over the streets makes all of it fast enough to route a person around the water in four milliseconds.

We have reported the failures beside the successes throughout, including three corrections against our own published claims, a negative result against 83 crowdsourced depth measurements, a drainage inversion that improved the quantity it was trained on and did nothing for the one that matters, and a learning loop whose gate has never fired because nobody has filed a report yet. We would rather the record be useful than flattering.

REPRODUCIBILITY

Code, all per-area artifact bundles including the Sentinel-1 water masks for the 15 areas that have them, road graphs, replay buffers and ranked site lists, together with 253 offline tests and step-by-step runbooks, are public at <https://github.com/Maverick-Ansh/project-varuna>. The live dashboard is at <https://project-varuna-iota.vercel.app> and its API at <https://anshvivek-varuna-floodtwin.hf.space>.

A new city builds with one script on a free GPU and serves from the committed bundle. Every headline number in this paper is regenerable from a committed artifact. The learned-baseline dataset is rebuilt by three builders, each with a verify mode that reports agreement with the committed copy rather than asserting it. The headline is unchanged on a fully rebuilt dataset, at 0.288 ± 0.008 against 0.288 ± 0.007 . The trivial-baseline builder reproduces all previously published values before extending them, which is how the fifteenth area was added to Table I without disturbing the other fourteen. Sentinel-1 masks for a new area are fetched by a script that needs one interactive Earth Engine login and ranks candidate dates by antecedent rainfall, so scenes are chosen for having weather in them rather than by hand.

ACKNOWLEDGMENT

This work uses FABDEM, ESA WorldCover, JRC Global Surface Water, Copernicus Sentinel-1, OpenStreetMap, Open-Meteo, NASA Black Marble, SoilGrids, HydroSHEDS and Central Ground Water Board data, all of which are free to use. The crowdsourced depth observations come from the public mumbaiiflood.in service. The system runs on free tiers provided by Google Colab, Kaggle, Hugging Face and Vercel. None of this work would be possible if those datasets and services were not open.

REFERENCES

- [1] P. D. Bates, M. S. Horritt, and T. J. Fewtrell, “A simple inertial formulation of the shallow water equations for efficient two-dimensional flood inundation modelling,” *Journal of Hydrology*, vol. 387, no. 1–2, pp. 33–45, 2010.
- [2] L. Hawker, P. Uhe, L. Paulo, J. Sosa, J. Savage, C. Sampson, and J. Neal, “A 30 m global map of elevation with forests and buildings removed,” *Environmental Research Letters*, vol. 17, no. 2, p. 024016, 2022.
- [3] D. Zanaga, R. Van De Kerchove, W. De Keersmaecker *et al.*, “ESA WorldCover 10m 2020 v100,” 2021, european Space Agency.
- [4] J.-F. Pekel, A. Cottam, N. Gorelick, and A. S. Belward, “High-resolution mapping of global surface water and its long-term changes,” *Nature*, vol. 540, pp. 418–422, 2016.
- [5] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- [6] J. Shaw, G. Kesserwani, J. Neal, P. Bates, and M. K. Sharifian, “LISFLOOD-FP 8.0: the new discontinuous Galerkin shallow-water solver for multi-core CPUs and GPUs,” *Geoscientific Model Development*, vol. 14, no. 6, pp. 3577–3602, 2021.
- [7] Z. Li, “Inunda: a GPU-native differentiable flood solver,” 2026, arXiv preprint arXiv:2607.09614. [Online]. Available: <https://arxiv.org/html/2607.09614>
- [8] Y. Liu *et al.*, “Universal shallow water equations with differentiable programming,” *Water Resources Research*, 2025.
- [9] “Toward AI-driven digital twins for metropolitan floods (CLDNet),” 2026, arXiv preprint arXiv:2605.13761. [Online]. Available: <https://arxiv.org/html/2605.13761v1>
- [10] “FlowsDT: a geospatial digital twin for navigating urban flood dynamics,” *Computers, Environment and Urban Systems*, 2026.
- [11] G. Nearing, D. Cohen, V. Dube, M. Gauch, O. Gilon, S. Harrigan, A. Hassidim, D. Klotz, F. Kratzert, A. Metzger *et al.*, “Global prediction of extreme floods in ungauged watersheds,” *Nature*, vol. 627, pp. 559–563, 2024.
- [12] R. Bentivoglio, E. Isufi, S. N. Jonkman, and R. Taormina, “Rapid spatio-temporal flood modelling via hydraulics-based graph neural networks,” *Hydrology and Earth System Sciences*, vol. 27, no. 23, pp. 4227–4246, 2023.
- [13] —, “Multi-scale hydraulic graph neural networks for flood modelling,” *Natural Hazards and Earth System Sciences*, vol. 25, no. 1, pp. 335–351, 2025.

- [14] A. Kazadi, J. Doss-Gollin, A. Sebastian, and A. Silva, "FloodGNN-GRU: a spatio-temporal graph neural network for flood prediction," *Environmental Data Science*, vol. 3, p. e21, 2024.
- [15] A. Kazadi, J. Doss-Gollin, and A. Silva, "Pluvial flood emulation with hydraulics-informed message passing," in *International Conference on Machine Learning (ICML)*, 2024.
- [16] "HydroGAT: distributed heterogeneous graph attention transformer for spatiotemporal flood prediction," 2025, arXiv preprint arXiv:2509.02481. [Online]. Available: <https://arxiv.org/pdf/2509.02481>
- [17] O. E. J. Wing, P. D. Bates, C. C. Sampson, A. M. Smith, K. A. Johnson, and T. A. Erickson, "Validation of a 30 m resolution flood hazard model of the conterminous United States," *Water Resources Research*, vol. 53, no. 9, pp. 7968–7986, 2017.
- [18] C. Marsh *et al.*, "Vertical accuracy of freely available global digital elevation models in flood-prone terrain," *International Journal of Digital Earth*, vol. 17, 2024.
- [19] T. H. Assumpção, I. Popescu, A. Jonoski, and D. P. Solomatine, "Citizen observations contributing to flood modelling: opportunities and challenges," *Hydrology and Earth System Sciences*, vol. 22, no. 2, pp. 1473–1489, 2018.
- [20] M. Mazzoleni, M. Verlaan, L. Alfonso, M. Monego, D. Norbiato, M. Ferri, and D. P. Solomatine, "Can assimilation of crowdsourced data in hydrological modelling improve flood prediction?" *Hydrology and Earth System Sciences*, vol. 21, no. 2, pp. 839–861, 2017.
- [21] S. Hesarkazzazi, A. E. Bakhshipour, M. Hajibabaei, U. Dittmer, A. Haghighi, and R. Sitzenfrei, "Generation of optimal (de)centralized layouts for urban drainage systems: a graph-theory combinatorial multi-objective optimisation," *Sustainable Cities and Society*, vol. 83, p. 103903, 2022.
- [22] "Respond to urban floods and groundwater depletion: a managed aquifer recharge approach," *Science of the Total Environment*, 2026.
- [23] P. Pavelic, K. Brindha, G. Amarnath, N. Eriyagama, L. Muthuwatta, and V. Smakhtin, "Controlling floods and droughts through underground storage: from concept to pilot implementation in the Ganges river basin," International Water Management Institute (IWMI), Tech. Rep. IWMI Research Report 165, 2015.
- [24] R. Torres, P. Snoeij, D. Geudtner *et al.*, "GMES Sentinel-1 mission," *Remote Sensing of Environment*, vol. 120, pp. 9–24, 2012.
- [25] P. Zippenfenig, "Open-Meteo.com weather API," 2023, open-source weather API with free access for non-commercial use.
- [26] OpenStreetMap contributors, "OpenStreetMap," 2026, data available under the Open Database License. [Online]. Available: <https://www.openstreetmap.org>
- [27] M. O. Román, Z. Wang, Q. Sun, V. Kalb, S. D. Miller, A. Molthan, L. Schultz, J. Bell, E. C. Stokes, B. Pandey, K. C. Seto *et al.*, "NASA's Black Marble nighttime lights product suite," *Remote Sensing of Environment*, vol. 210, pp. 113–143, 2018.
- [28] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *International Conference on Machine Learning (ICML)*, 2017, pp. 1263–1272.
- [29] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, "FiLM: Visual reasoning with a general conditioning layer," in *AAAI Conference on Artificial Intelligence*, 2018.